

Kochenov Geometric Identity for G

An Empirical Observation

Boris V. Kochenov
kochenovforwork@gmail.com
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I. Main Result

The gravitational constant admits an exact geometric representation:

$$G \times 10^{11} = 31^2 \times \sqrt{5} / \phi^{12} = 6.67354674496161812375... \quad (1)$$

Source	Value (x10 ⁻¹¹)	Gap from Gk
Schlaminger (2014)	6.6729 +/- 0.0005	+0.010%
Gk (this work)	6.67354674496...	--
CODATA 2018	6.67430 +/- 0.00015	-0.011%
Labs disagree		0.021%

Gk lies between both measurements. Each gap is smaller than the mutual disagreement between laboratories. Inputs: phi, 31, sqrt(5) only. No measured constants. No fitting.

II. The Generator Identity

The expression $\phi^{12}/\sqrt{5}$ is the Binet approximation to $F_{12} = 144$, the unique perfect square in the Fibonacci sequence. Direct evaluation yields:

$$\phi^{12} / \sqrt{5} = 144 + 1/720 + O(10^{-8}) \quad (2)$$

where $720 = 6!$ -- a unique intersection of the exponential sequence (ϕ^n), the factorial sequence ($m!$), and Fibonacci numbers (F_n) at indices $n = 12$, $m = 6$. No comparable alignment exists for nearby indices. The absolute deviation is $|\Delta| = 1.39 \times 10^{-8}$, with relative precision $|\Delta|/144 = 9.7 \times 10^{-11}$.

III. The Residual 1/720 -- Three Independent Domains

The same residual 1/720 appears independently in three physical domains:

Domain	Formula	Gap
Geometry	$\phi^{12}/\sqrt{5} - 144 = 1/720$	9×10^{-9}
Nuclear	$mn = mp \times (1 + 1/720)$	-0.001%
EM	$\alpha^{-1} = 144 - 7 - 1/720$	-0.027%

Three independent physical domains. One residual. The geometry identity is exact to 10^{-8} . The nuclear result (neutron-to-proton mass ratio) is verified against CODATA to 0.001%. The electromagnetic result uses the rounded value $\alpha^{-1} = 137$.

IV. Open Question

The integer $31 = 2^5 - 1$ (Mersenne prime) is the structural anchor of the formula. Its geometric origin remains under investigation.

V. What This Is Not

This is an empirical observation. No dynamical model is claimed. No CODATA values were used in the derivation. The formula is presented as a geometric identity whose physical significance, if any, remains to be established.

Full precision reference:

Gk = 6.67354674496161812375919977179042360791321745487151556
55761367183204332559409890102447913340675769273 $\times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$

Derived from: $Gk = 31^2 \times \sqrt{5} / (\phi^{12} \times 10^{11})$